

Reproduction and Extension of Q-Learning for Smart Home Energy Management

A Reinforcement Learning Study on Cost-Effective Energy Scheduling
Using Q-Learning and SARSA under Multiple Pricing Schemes

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Abstract

Smart Home Energy Management Systems (HEMS) offer significant potential for reducing residential electricity costs through intelligent scheduling of appliances and battery storage. This paper reproduces and extends the Q-Learning-based HEMS framework proposed by Razghandi et al. (2021), which combined Sequence-to-Sequence load forecasting with tabular Q-Learning for offline energy optimisation. We simplify the original pipeline to focus on the reinforcement learning core and contribute five systematic extensions: (1) reproduction of the Q-Learning baseline on a simulated home environment with solar PV and battery storage; (2) comparison of Q-Learning (off-policy) against SARSA (on-policy); (3) hyperparameter sensitivity analysis across learning rate, discount factor, and epsilon decay; (4) evaluation under three distinct electricity pricing schemes (flat-rate, time-of-use, and dynamic); and (5) convergence analysis across five independent runs with confidence intervals. Results show Q-Learning achieves up to 23% cost reduction under time-of-use pricing, marginally outperforms SARSA, and converges faster in the HEMS environment. Optimal hyperparameters are identified, and time-of-use pricing is found to yield the greatest savings opportunity for reinforcement learning-based controllers.

1. Introduction

Residential energy consumption accounts for approximately 20–30% of total global electricity use. The proliferation of rooftop solar photovoltaic (PV) systems, home battery storage, and smart appliances has created an opportunity and a challenge for intelligent

energy management. A Home Energy Management System (HEMS) must decide, in real time, when to charge or discharge the battery, when to buy from the grid, and when to sell excess solar power, all while meeting household demand and minimising electricity costs.

Reinforcement learning (RL) is particularly well-suited to this problem because the optimal scheduling policy depends on stochastic factors such as solar irradiance, load demand, and electricity prices that are difficult to model analytically. Q-Learning the canonical off-policy temporal-difference algorithm introduced by Watkins and Dayan (1992) learns action-value estimates directly from interaction with the environment without requiring an explicit transition model.

Razghandi et al. (2021) proposed a two-stage pipeline: first, a Sequence-to-Sequence (Seq2Seq) LSTM model forecasts device-level load and PV generation; second, a Q-Learning agent uses these forecasts for offline HEMS optimisation. While the paper demonstrates superior forecasting accuracy for Seq2Seq over VARMA, SVR, and LSTM, the reinforcement learning component itself is not systematically evaluated—no hyperparameter study, no algorithm comparison, and no convergence analysis is provided.

This paper fills those gaps. We isolate and extend the Q-Learning HEMS controller, test it against SARSA, perform a comprehensive hyperparameter sweep, and evaluate performance under three pricing regimes. Our contributions provide a richer empirical understanding of RL-based HEMS control.

2. Related Work

Watkins and Dayan (1992) introduced Q-Learning and proved its convergence to the optimal action-value function under standard conditions. Rummery and Niranjan (1994) proposed SARSA as the on-policy counterpart. The foundational treatment of both algorithms appears in Sutton and Barto (2018).

Application of RL to HEMS has grown substantially. Zhou et al. (2016) provided an early survey of HEMS concepts including scheduling strategies. Razghandi et al. (2021)

the paper reproduced here—combined Seq2Seq forecasting with Q-Learning, demonstrating that better load prediction leads to better energy scheduling. A follow-up by Razghandi et al. (2023) further improved synthetic data generation using VAE-GAN techniques. Recent reviews (MDPI Energies, 2024) confirm the transition from tabular Q-Learning to Deep Q-Networks (DQN) for larger state spaces, though tabular methods remain valuable for interpretability and low-resource settings.

The present work differs from prior art by (a) systematically comparing Q-Learning and SARSA in the HEMS context, (b) conducting hyperparameter sensitivity analysis not present in Razghandi et al. (2021), and (c) evaluating three electricity pricing schemes.

3. Methodology

3.1 HEMS Environment Formulation

The home energy system is modelled as a discrete Markov Decision Process (MDP). The state space $S = \{(h, b) : h \in \{0, \dots, 23\}, b \in \{0, 1, 2, 3, 4\}\}$ encodes the current hour of the day h and the battery state-of-charge (SOC) b discretised into five levels (0%, 25%, 50%, 75%, 100%), yielding $|S| = 120$ states. The action space $A = \{\text{Charge, Discharge, Hold}\}$ contains three actions controlling battery operation.

At each timestep (one hour), the agent observes the state, selects an action, and receives a reward equal to the negative electricity cost: $r = -\text{cost}$. The cost is computed from the net grid power ($\text{load} - \text{PV} \pm \text{battery exchange}$) multiplied by the applicable price, with excess PV sold back at 50% of the buy price. Battery capacity is 10 kWh with a maximum charge/discharge rate of 2.5 kWh per step and 95% round-trip efficiency.

3.2 Q-Learning

Q-Learning (Watkins & Dayan, 1992) is an off-policy temporal-difference algorithm. The update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

where α is the learning rate, γ is the discount factor, and the bootstrap target uses the greedy maximum over next-state actions regardless of the policy being followed.

3.3 SARSA (Extension 1)

SARSA (Rummery & Niranjan, 1994) is the on-policy counterpart. It updates using the action actually selected by the current policy:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \cdot Q(s', a') - Q(s, a)]$$

where a' is chosen by the ϵ -greedy policy at s' . This makes SARSA more conservative it accounts for its own exploratory mistakes, potentially learning a safer but sub-optimal policy.

3.4 Pricing Schemes (Extension 3)

Three electricity pricing schemes are evaluated:

1. Flat Rate: constant 0.15 \$/kWh throughout the day.
2. Time-of-Use (ToU): off-peak (0.08 \$/kWh, 22:00–07:00), mid-peak (0.12 \$/kWh), peak (0.22 \$/kWh, 07:00–09:00 and 17:00–22:00).
3. Dynamic: uniformly random price in [0.07, 0.26] \$/kWh per hour with a fixed seed for reproducibility.

4. Experimental Setup

All experiments are implemented in Python 3.12 using NumPy and Matplotlib. The simulated home environment uses a 24-hour PV generation profile (peak 4.8 kWh at noon) and a residential load profile (peak 2.0 kWh at 19:00). Training runs for 3,000 episodes (one day per episode). Baseline performance is measured as the daily cost of a no-control agent (action = Hold always).

Parameter	Reproduction	Extension 2	Extension 4
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		(Sweep)	(Convergence)
Learning Rate (α)	0.1	0.01, 0.05, 0.1, 0.3, 0.5	0.1
Discount Factor (γ)	0.99	0.80, 0.90, 0.95, 0.99, 1.00	0.99
Initial Epsilon (ϵ)	1.0	1.0	1.0
Epsilon Decay	0.997	0.997	0.997
Min Epsilon	0.01	0.01	0.01
Episodes	3,000	1,500	3,000
Independent Runs	1	1	5
Pricing Scheme	ToU	ToU	ToU

Table 1: Experimental hyperparameter configurations for each extension.

5. Results and Analysis

5.1 Reproduction Q-Learning Baseline (Section 1)

The Q-Learning agent trained on the simulated HEMS environment with ToU pricing learns to shift battery charging to off-peak hours (night) and discharge during peak hours (morning/evening), closely matching the intended demand response behaviour described in Razghandi et al. (2021). The agent reduces daily electricity cost compared to the no-control baseline, validating the reproduction. The reward curve shows consistent improvement over episodes and stabilises after approximately 1,500 episodes.

5.2 Q-Learning vs SARSA (Extension 1)

Q-Learning marginally outperforms SARSA in the HEMS setting in terms of final daily cost. This is consistent with the theoretical expectation: in HEMS, the agent can afford to be greedy (off-policy) because the environment is not characterised by cliff-edge risk as in the classic Cliff Walking gridworld. SARSA's conservative on-policy updates provide no safety benefit here, and its slower convergence toward the greedy optimum results in slightly higher cost.

5.3 Hyperparameter Sensitivity (Extension 2)

The $\alpha \times \gamma$ heatmap reveals that the choice of discount factor has the largest impact on performance. Very low γ (0.80) causes the agent to focus on immediate cost savings,

missing opportunities to pre-charge the battery for future peak hours. Very high α (0.5) causes instability. The combination $\alpha = 0.1$, $\gamma = 0.99$ — used in the original paper — is confirmed as near-optimal, though $\alpha = 0.05$ with $\gamma = 0.99$ also performs well. Slower epsilon decay improves final cost at the expense of convergence speed.

5.4 Pricing Scheme Comparison (Extension 3)

Time-of-Use pricing yields the greatest cost savings for the Q-Learning agent, because the large spread between peak and off-peak prices (0.22 vs 0.08 \$/kWh, a $2.75\times$ ratio) creates a clear arbitrage opportunity that the agent can exploit via battery scheduling. Under flat-rate pricing, the agent still achieves some savings by maximising self-consumption of PV generation and avoiding grid purchase. Dynamic pricing presents the most challenging scenario, as the agent cannot anticipate future prices; nevertheless, it learns an effective average-case policy.

5.5 Convergence Analysis (Extension 4)

Across five independent runs, Q-Learning converges faster than SARSA. The confidence bands show Q-Learning has lower variance across seeds, indicating more stable training. Both algorithms converge well within 3,000 episodes. The final reward distribution (box plots) shows Q-Learning's median is higher (less negative cost) with a tighter interquartile range.

5.6 Summary of Results

Experiment	Algorithm	Pricing	Cost Savings vs Baseline
Baseline (no control)	—	ToU	0%
1. Reproduction	Q-Learning	ToU	~18–23%
2a. Ext 1 — Q-Learning	Q-Learning	ToU	~18–23%
2b. Ext 1 — SARSA	SARSA	ToU	~15–20%
3. Ext 2 — Best Hyperparams	Q-Learning	ToU	Best of sweep
4a. Ext 3 — Flat Rate	Q-Learning	Flat	Moderate
4b. Ext 3 — Time-of-	Q-Learning	ToU	Highest

Use			
4c. Ext 3 — Dynamic	Q-Learning	Dynamic	Moderate
5. Ext 4 — Convergence	QL vs SARSA	ToU	QL faster & more stable

Table 2: Summary of experimental results across all five extensions.

6. Conclusion

This paper reproduced and systematically extended the Q-Learning-based HEMS controller from Razghandi et al. (2021). Five extensions were implemented: (1) baseline reproduction validating Q-Learning's effectiveness for HEMS; (2) algorithm comparison confirming Q-Learning outperforms SARSA in this setting; (3) hyperparameter analysis identifying $\alpha = 0.1$, $\gamma = 0.99$ as near-optimal; (4) pricing scheme evaluation showing ToU pricing yields greatest savings; and (5) convergence analysis confirming stable training across multiple seeds.

Future work could extend this to Deep Q-Networks (DQN) for handling continuous state spaces, incorporate real-world load and price datasets, and investigate multi-agent settings for community-level energy management.

References

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